



Marketing Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Unraveling Multifaceted User Preferences on Digital Platforms: A Bayesian Deep-Learning Approach

Mingzhang Yin, Ziwei Cong, Jia Liu

To cite this article:

Mingzhang Yin, Ziwei Cong, Jia Liu (2026) Unraveling Multifaceted User Preferences on Digital Platforms: A Bayesian Deep-Learning Approach. Marketing Science

Published online in Articles in Advance 21 Apr 2026

. <https://doi.org/10.1287/mksc.2024.1090>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2026, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Unraveling Multifaceted User Preferences on Digital Platforms: A Bayesian Deep-Learning Approach

Mingzhang Yin,^a Ziwei Cong,^{b,*} Jia Liu^c

^aDepartment of Marketing, University of Florida, Gainesville, Florida 32611; ^bDepartment of Marketing, Georgetown University, Washington, District of Columbia 20057; ^cDepartment of Marketing, IEDA, Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong, China

*Corresponding author

Contact: mingzhang.yin@warrington.ufl.edu,  <https://orcid.org/0000-0002-5216-2437> (MY); zc260@georgetown.edu,  <https://orcid.org/0000-0001-5196-7527> (ZC); jialiu@ust.hk,  <https://orcid.org/0000-0002-0279-724X> (JL)

Received: September 20, 2024

Revised: September 13, 2025;
January 26, 2026

Accepted: February 1, 2026

Published Online in Articles in Advance:
April 21, 2026

<https://doi.org/10.1287/mksc.2024.1090>

Copyright: © 2026 INFORMS

Abstract. Given the increasing importance of user engagement on digital platforms, this paper proposes a Bayesian deep-learning model called the Multi-Dynamic Neural Poisson System (MDNPS), which can capture the multifaceted nature of user sparse but high-dimensional activities on such platforms. MDNPS yields semantically interpretable factors underlying high-dimensional items, quantifies user preferences at different granularity levels, and adeptly captures their temporal and cross-activity dependencies. This model is scalable to large empirical data and can be inferred efficiently with a stochastic variational Bayes algorithm. We apply MDNPS to the largest knowledge-sharing platform in China, focusing on the dynamics in user content consumption and contribution behaviors. We show that MDNPS significantly outperforms benchmark factor models in factor quality and out-of-sample data fitting. Our platform-level and individual-level estimates show rich and interpretable insights about consumer preference dynamics and their relationships with user popularity. These insights offer valuable managerial implications for platforms to understand user segments, personalize content offerings, and enhance user engagement.

History: Tat Chan served as the senior editor for this article.

Funding: This work was supported by the University Grants Committee of the Hong Kong Special Administrative Region (Project No. 16500124, 2024-27). All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or nonfinancial interest in the subject matter or materials discussed in this manuscript. M. Yin acknowledges the support from the Marketing Science Institute and the Warrington Commitment Research Award. The authors acknowledge the computational support from UF Research Computing and HiPerGator.

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mksc.2024.1090>.

Keywords: digital platforms • dynamic model • neural networks • preference measurement • user-generated content

1. Introduction

With the advent of digital technology, online platforms have emerged as primary venues for consumers to interact with products, services, and content. For instance, 71% of U.S. consumers get news content via social media, 84% engage in online communities to share their experiences and hobbies, and 58% turn to the Internet to seek customized solutions (Horrigan 2001, Fox 2007). This digital transformation has generated vast, high-dimensional behavioral data. These data hold immense potential for understanding user preferences and guiding strategic decisions, such as recommendation, targeted advertising, and trend forecasting (Hemann and Burbary 2013). The sheer scale and complexity of these data have led researchers and practitioners to rely heavily on black-box algorithms.

Although these algorithms offer predictive power with flexible modeling, they obscure the underlying behavioral patterns, making it difficult to interpret user preferences or design transparent and accountable platform policies. As platforms grow and diversify, the need for interpretable models becomes more urgent—not only to improve algorithmic decisions but also to support strategic policies that go beyond automation. For example, platforms must decide how to allocate resources for content creation, identify emerging areas of user interest, and segment users based on evolving preferences. These decisions require visibility into the structure and dynamics of user behavior, which black-box algorithms cannot provide.

However, extracting interpretable and meaningful insights about user preferences involves several

challenges that extant models cannot address simultaneously. Many digital platforms, especially social media and content-oriented ones, present an almost infinite and ever-increasing number of alternatives (Dhillon and Aral 2021). Yet individual users engage with only a small subset, which requires efficient dimension-reduction techniques. More importantly, users' preferences on digital platforms are highly heterogeneous across multiple dimensions, which calls for flexible modeling structures. Users' behavior is shaped by external influences, such as platform-wide social trends, as well as by personal experiences. Their preferences are activity specific (Cain 2022) and evolve over time (Dew et al. 2020). For example, a user actively *generating* science and technology content may be interested in *consuming* content about Bitcoin; the user's interactions with Bitcoin content are influenced not only by personal taste but also by short-term trends and long-term seasonality on the platform. Thus, a joint model capturing multifaceted user behaviors is crucial for preference measurement with consistent interpretation and temporal continuity.

This paper addresses these challenges by introducing a multi-dynamic neural Poisson system (MDNPS)—a novel Bayesian deep-learning framework that infers interpretable, multifaceted user preferences from high-dimensional time series tensors. MDNPS integrates the transparency of probabilistic modeling (e.g., Bayesian methods) with the flexibility of deep neural networks (Wang and Yeung 2020), enabling scalable inference while preserving interpretability.¹ Specifically, MDNPS utilizes a hierarchical framework to model preference dynamics at both the platform and individual levels. The uncovered preferences reveal heterogeneity across multiple correlated activities and exhibit structured, continuous temporal evolution. At the platform level, it uses long short-term memory (LSTM) networks to capture complex short-term and long-term trends. At the individual level, it employs a correlated hidden Markov model (HMM) to capture cross-activity interactions, where one activity is influenced by all the activities at a previous time step. A stochastic variational Bayes algorithm is developed to enable efficient and scalable Bayesian inference. These characteristics make the MDNPS suitable for various settings that involve multiple large-scale, high-dimensional, and dynamic user activities (e.g., interactions with products, services, text, and images). To the best of our knowledge, this paper is among the first to apply Bayesian deep learning for preference measurement.

We demonstrate the value of MDNPS using longitudinal data from China's largest online knowledge-sharing platform, where users both produce and consume content in a dynamic environment shaped by individual interests and platform-wide trends. We apply MDNPS to several thousand active users from

2011 to 2017, focusing on answering questions (i.e., reflecting user contribution preferences) and subscribing to questions (i.e., reflecting user consumption preferences). We show that MDNPS achieves superior data fitting compared with a broad set of static and dynamic benchmark models. More importantly, MDNPS yields interpretable representations of how user interests evolve at both the individual and platform levels. That is, unlike purely black-box models that obscure the sources of behavioral change, MDNPS explicitly recovers latent preference trajectories and activity-specific engagement patterns that align with intuitive notions of learning, exploration, and shifting attention. The model thus functions as a behavioral mining tool, translating complex user histories into structured and economically meaningful signals rather than opaque embeddings.

By compressing complex, rich activity streams into interpretable summaries, MDNPS enables a wide range of downstream analyses that are difficult to support with black-box approaches, including segmenting users by preference dynamics, diagnosing divergences between consumption and contribution, and distinguishing trend-driven behavior from intrinsic motivation. These capabilities enable platforms to move beyond reactive algorithmic decisions toward strategic content and engagement policies. As a proof of concept, we further show that the interpretable embeddings generated by MDNPS can be leveraged directly for content recommendation, producing transparent, activity-specific suggestions even with sparse tag-level information. Together, these applications highlight MDNPS as a scalable and flexible framework that combines strong empirical performance with interpretability, enabling data-driven yet transparent decision-making for digital platforms.

The rest of the paper is organized as follows. We first review relevant literature. Then, we introduce MDNPS and its inference algorithm, followed by the description of our empirical application. We present and discuss various substantive outputs of our application. Finally, we conclude with a discussion.

2. Related Literature

This paper contributes to the literature on modeling consumer preferences. The classic approach by Guadagni and Little (1983) captures brand preferences using exponential smoothing of customer-level brand loyalty parameters. Subsequent studies have incorporated product interdependencies (Duvvuri et al. 2007) and interactions between consumers (Yang and Allenby 2003). More recent work has focused on estimating dynamic user preferences for physical products through a single kind of consumer activity (Dew et al. 2020, Li and Ma 2020, Dhillon and Aral 2021, Jacobs

et al. 2021, Liu and Dzyabura 2021, Yin et al. 2025). Among them, Liu and Dzyabura (2021) allowed multiple partworth vectors associated with each individual when specifying the purchase utility. Our modeling framework advances this literature by capturing preferences on latent factors (rather than observed items) while allowing for simultaneous temporal and cross-activity dependencies.

Our proposed method builds on the literature of factor models, particularly Poisson factor analysis (PFA), a model widely used for high-dimensional, sparse count data in marketing (Liu et al. 2021, 2024; Toubia 2021). For time-series count data, Schein et al. (2016) introduced the Poisson-Gamma Dynamical Systems (PGDS) to capture the short-term evolution using a hidden Markov process, whereas subsequent works imposed hierarchical models to account for long-term historical influences on topic intensities (Gong and Huang 2017, Guo et al. 2018). Our MDNPS extends this literature by integrating deep neural networks and Markov models in jointly modeling long-term, short-term, and cross-activity interdependencies.

Our research is also related to the growing applications of deep-learning methods in marketing problems (Liu 2023, Malhotra et al. 2023). Recurrent neural networks (RNNs) such as the LSTM (Yu et al. 2019) have been used for predictive tasks (Liu et al. 2019, Chakraborty et al. 2022); similarly, the Transformer model has been applied to predict customer journey (Yin et al. 2024b, Lu and Kannan 2026). Despite their predictive strength, black-box sequential models often lack interpretable behavioral structure and rely on heuristic regularization (Sisodia et al. 2025) or post hoc feature engineering (Cheng et al. 2022). Our modeling framework advances such applications along two dimensions. First, we focus on overcoming sparsity problems and offering interpretable measurements of user preferences by combining LSTM with transparent factor models. Second, we use the LSTM to improve the flexibility of the prior and posterior distributions in a Bayesian framework rather than a predictive architecture.

Our application context is related to the literature on user interest dynamics on digital content platforms. Previous studies have focused primarily on dynamics driven by short-term external events or platform interventions (Dhillon and Aral 2021, West et al. 2021, Liberali and Ferecatu 2022, Liu and Cong 2023, Yin et al. 2024a). We complement this literature by considering users' long-run evolution throughout their lifetime on content platforms, which generates important managerial implications for dynamic targeting and segmentation. In the context of knowledge-sharing platforms, our work enriches the understanding of user online learning and self-improvement (Koksal 2020, Goli et al. 2022). Unlike traditional education research that relies on cross-sectional data, we explore within-individual

learning trajectories using granular data. Our model and empirical findings can be useful for many online knowledge-sharing platforms (e.g., Quora, Coursera, Reddit, edX, Udacity) to improve user learning experiences.

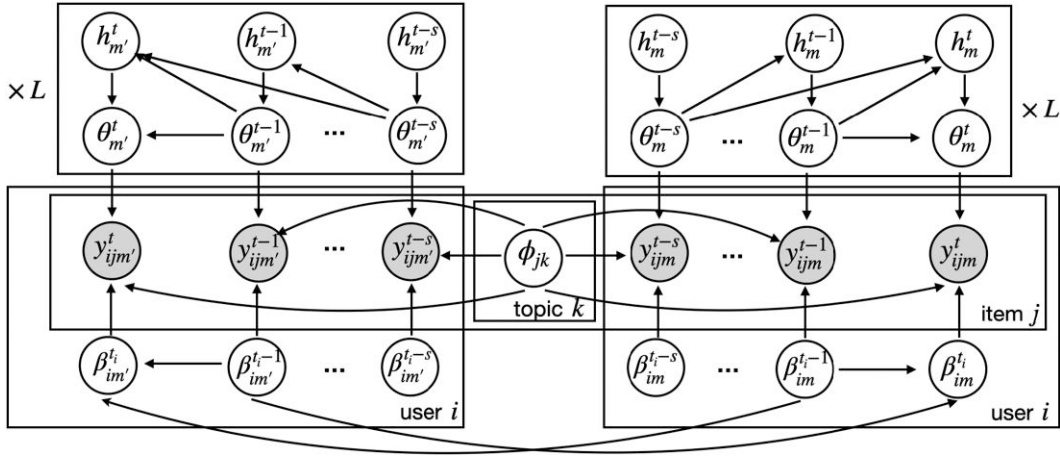
3. Multi-Dynamic Neural Poisson System

Our proposed MDNPS is a deep dynamic factor model that captures the evolution of user multifaceted preferences on digital platforms, where users often dynamically interact with many items (e.g., tags, products, images, and users) in multiple ways (e.g., following, subscribing, liking, purchasing, and answering). Our model design is guided by several desiderata for understanding users' multifaceted preferences on digital platforms. First, to generate actionable insights for content management, the model should summarize users' sparse interactions over large-scale items into low-dimensional, interpretable latent factors. Second, it is important to disentangle population-level preferences driven by platform-wide trends from idiosyncratic heterogeneity shaped by individual tastes. Third, the model should account for interdependence across multiple user actions that influence each other. Lastly, it should capture both short-term fluctuations and long-run dynamics in preference evolution revealed through these actions. In the following, we first present our model specification that addresses these considerations and then introduce an efficient inference algorithm.

3.1. Model Specification

Let the corpus consist of N users, J items, M types of user activities, and T periods. The observed data are a three-dimensional dynamic tensor. At each time t , we observe the activity count of user i interacting with item j using action type m , denoted by y_{ijm}^t for $i \in \{1, \dots, N\}$, $j \in \{1, \dots, J\}$, and $m \in \{1, \dots, M\}$. Figure 1 presents the graphical model of the MDNPS, where shared nodes represent the data and unshared nodes represent the model parameters.

3.1.1. Summarizing Sparse Interactions as Topics. In practice, both N and J are typically large, but most users interact with only a very small subset of items, resulting in sparse user-item interactions. This large-scale yet sparse data structure complicates item-level analysis, motivating us to adopt factor models, particularly the Poisson factorization, for dimension reduction. Accordingly, we reduce the dimensionality of these actions by projecting the items into K latent topics, where $K \ll J$. Suppose the K topics (a.k.a. factors) are represented by the time-invariant $J \times K$ matrix Φ . Each $J \times 1$ vector Φ_k represents the item distribution of topic k .

Figure 1. Graphical Model of the MDNPS

Notes. The subscripts m and m' represent distinctive user activities. The superscript is the user's relative lifetime since joining the platform for β and is the calendar time for other variables.

We specify its prior distribution on the logit space as a standard Normal distribution:

$$\tilde{\phi}_k \sim \mathcal{N}(0, I) \quad \text{and} \quad \phi_k = \text{softmax}(\tilde{\phi}_k). \quad (1)$$

This prior distribution is equivalent to a standard logistic normal distribution for ϕ_k (Atchison and Shen 1980). The softmax function $\text{softmax}(\phi) = \exp(\phi) / \sum_{j=1}^J \exp(\phi_j)$ makes ϕ_k a valid item distribution for the topic k with $\sum_{j=1}^J \phi_{jk} = 1$. Thus, each topic can be interpreted based on the items with the highest probabilities. Note that the same latent topics are shared across periods and activities, ensuring a consistent topic definition and enabling analysis of how preferences evolve and interrelate across these topics. This approach attributes individual activities to evolving preferences or trends rather than to changing topic definitions, thereby enabling us to distinguish user preference dynamics from shifts in topic components.

3.1.2. Decomposing Population and Individual Preferences.

We model the individual activity count y_{ijm}^t using user topic-specific preferences while decomposing it into platform and individual levels. Specifically, we assume that y_{ijm}^t follows a Poisson distribution with a continuous rate parameter $\mu_{ijm}^t > 0$,

$$y_{ijm}^t \sim \text{Pois}(\mu_{ijm}^t). \quad (2)$$

Because users enter digital platforms at different times, we do not observe the activities of user i for all the time $t \in \{1, \dots, T\}$. Thus, we let t_i^0 denote the time that user i joined the platform and set $t_i = t - t_i^0$ as user i 's lifetime on the platform. We then define the user-specific parameters based on the user's lifetime rather than the calendar time. The lifetime t_i allows for individual-specific lifecycle on the platform despite receiving the same platform-level shocks at a given time t .

We decompose the Poisson rate parameter as a sum across the K topics,

$$\mu_{ijm}^t = \sum_{k=1}^K \theta_{mk}^t \beta_{imk}^{t_i} \phi_{jk}. \quad (3)$$

where $\phi_k \in \mathbb{R}^J$ is the item distribution of topic k , summing to one, θ_{mk}^t captures population preferences, and $\beta_{imk}^{t_i}$ captures user i 's preferences at lifetime t_i . This formulation means that a user's engagement with an item is shaped by the joint influence of the platform-wide interest in the topic (θ) and that user's personal interest in it (β), weighted by how strongly the item represents that topic (ϕ). The platform-wide θ_{mk}^t is often influenced by major external events and trends, whereas $\beta_{imk}^{t_i}$ reflects how an individual's interests evolve along with the aggregate trends, akin to individual random effects. If θ_{mk}^t is high but $\beta_{imk}^{t_i}$ is low, the user's behavior is likely trend driven; if $\beta_{imk}^{t_i}$ is high but θ_{mk}^t is low, the user is likely pursuing a niche interest.

We use multiplication between θ_{mk}^t and $\beta_{imk}^{t_i}$ as a result of several considerations. The rate of Poisson distribution being positive necessitates positivity of $(\theta_{mk}^t, \beta_{imk}^{t_i}, \phi_{jk})$, and if the interaction is defined additively as $\theta_{mk}^t + \beta_{imk}^{t_i}$, then the individual preference can only amplify the population trends because $\theta_{mk}^t + \beta_{imk}^{t_i} \geq \theta_{mk}^t$. This would estimate population trends inaccurately because the activity strength of an individual is not always stronger than that of the population. In contrast, in Equation (3), user i 's taste can be stronger than the platform trend when $\beta_{imk}^{t_i}$ is larger than one or weaker when $\beta_{imk}^{t_i}$ is less than one.

In summary, our factor model adopts a hierarchical structure to disentangle different factors driving users' topical preferences over time, which allows for flexible dynamic modeling. As we will describe below, this

hierarchy of platform and individual levels enables the model to capture complex temporal dependencies in the evolution of platform trends (θ_{mk}^t) while capturing cross-activity dependence in the evolution of individual tastes (β_{imk}^t). This model design balances flexibility and computational efficiency.

3.1.3. Capturing Platform-Level Dynamics and Long-Term Dependence. We now describe the evolution of platform-level topic strength θ_{mk}^t and model its temporal variations with flexible neural network methods. In general, the dynamic patterns of topics at the platform level are often complex. For example, on a Q&A platform, a political campaign may drastically increase user interest in political content across a few months of the election season, which may also impact the popularity of political content in future election cycles, resulting in a long-term dependency. Furthermore, the popularity of a topic like “E-commerce” may depend on the dynamics of related topics (e.g., “Digital Media” and “Online Communities”), and this dependency is generally nonlinear. Given the complex cross-time and cross-topic dependence, we leverage the LSTM neural network (Hochreiter and Schmidhuber 1997) to model the evolution of the platform-level topic strength θ flexibly.

For ease of notation, we omit the subscript m in θ_{mk}^t when describing the LSTM specification because the structure and process are the same across activity types. We use $\theta^t \in \mathbb{R}^K$ to denote the vector $(\theta_1^t, \dots, \theta_K^t)^\top$. We first remove the positivity restriction by projecting each θ^t to the logarithmic space using an embedding matrix $E \in \mathbb{R}^{d_h \times d_h}$ as $\hat{\theta}^t = E \ln(\theta^t)$, $\hat{\theta}^t \in \mathbb{R}^{d_h}$ with dimension d_h ; this allows an unrestricted input to the LSTM. Then, we transform the sequence of embeddings $\{\hat{\theta}^0, \hat{\theta}^1, \dots, \hat{\theta}^{T-1}\}$ to a sequence of hidden states $\{h^1, h^2, \dots, h^T\}$, $h^t \in \mathbb{R}^{d_h}$ by the LSTM, represented as a mapping $f(\cdot): \mathbb{R}^{d_h \times T} \rightarrow \mathbb{R}^{d_h \times T}$,

$$(h^1, h^2, \dots, h^T) = f(\hat{\theta}^0, \hat{\theta}^1, \dots, \hat{\theta}^{T-1}), \quad (4)$$

where $\hat{\theta}^0 \sim \mathcal{N}(0, \mathbf{I}_K)$ is a padding vector. This LSTM mapping is designed to capture long-term dependence in platform trends such that each θ^t depends on all $\theta^{1:t-1}$ in the previous time steps. This is an essential desiderate in modeling population dynamics, such as in our political campaign example.

The LSTM mapping $f(\cdot)$ consists of a set of functions (r_t, i_t, o_t) that are standard LSTM modules (Yu et al. 2019). For completeness, these are a forget gate r_t ($r_t = \sigma(W_f \cdot [h^{t-1}, \hat{\theta}^{t-1}] + b_f)$, $\sigma(z) = 1/(1 + \exp(-z))$), an input gate i_t ($i_t = \sigma(W_i \cdot [h^{t-1}, \hat{\theta}^{t-1}] + b_i)$), and an output gate o_t ($o_t = \sigma(W_o \cdot [h^{t-1}, \hat{\theta}^{t-1}] + b_o)$), which enable the function f to store important information and forget

irrelevant information in a long sequence (see Online Appendix A.5 for details on our use of the LSTM). The training parameters of the LSTM are the weight matrices W_f, W_i, W_o and bias vectors b_f, b_i, b_o of these functions. Note that in this transformation, each latent vector h^t depends on all the preceding $\hat{\theta}^{0:t-1}$, and we denote this long-term dependency explicitly as $h^t(\hat{\theta}^{0:t-1})$.

With the hidden states $\{h^1, h^2, \dots, h^T\}$, the information before time t propagates to θ^t by

$$p(\theta^t | \theta^{1:t-1}) = \mathcal{LN}(m(\hat{\theta}^{t-1}, h^t(\hat{\theta}^{0:t-1})), \text{diag}(\sigma^2(\hat{\theta}^{t-1}, h^t(\hat{\theta}^{0:t-1}))). \quad (5)$$

This distribution indicates that the platform trends θ^t at time t depend on the entire history of trends from steps 1 through $t-1$ rather than solely on the immediately preceding step. This long-term pattern is modeled flexibly through neural networks. In Equation (5), $m(\cdot)$ and $\sigma(\cdot)$ are mean and variance vectors of the platform parameter θ^t at time t . They are modeled as two separate feed-forward neural networks with parameters W_μ and W_σ , which take the concatenated $(\hat{\theta}^{t-1}, h^t)$ as the inputs, and output the K -dimensional mean and variance vectors, respectively. $\text{diag}(\sigma^2)$ denotes a diagonal covariance matrix whose diagonal entries are the squared outputs of $\sigma(\cdot)$, and $\mathcal{LN}(m, \text{diag}(\sigma^2))$ is the log-normal distribution that ensures the positivity of θ^t . We let W_θ denote all the parameters involved in this part, including the embedding matrix E , the neural network parameters $\{W_f, W_i, W_o, b_f, b_i, b_o\}$ of the LSTM, and the parameters $\theta_\mu, \theta_\sigma$ of the feedforward neural networks $m(\cdot)$ and $\sigma(\cdot)$.²

3.1.4. Capturing Individual-Level Dynamics and Cross-Activity Dependence. We finally model the evolution of individual-level offset β_{imk}^t and explicitly encode the interplay between multiple user actions on a platform. Because the number of individual parameters is large and the individual-level data are sparse, we use a transparent hidden Markov approach instead of the LSTM to model β_{imk}^t for computational efficiency and accuracy. We assume that the K -dimensional vector $\beta_{im}^t := (\beta_{im1}^t, \dots, \beta_{imK}^t)^\top$ is influenced by user i 's preferences for all the activity types at the previous time step $t-1$. In other words, we allow individual users' different types of preferences to influence each other over time.

To this end, we specify β_{im}^t using a correlated Markov model with transition matrices $\Pi^{m'm}$ as

$$p(\beta_{im}^t | \beta_{i,1:M}^{t-1}) = \mathcal{LN}\left(\sum_{m'=1}^M \Pi^{m'm} \ln(\beta_{im'}^{t-1}), \sigma_0^2 \mathbf{I}\right). \quad (6)$$

The dynamic transition allows the preference via activity m to be directly related to all the activity types $m' = 1, \dots, M$ at a previous time. This cross-activity

dependency is essential on two-sided platforms like a Q&A platform, for example, because more question subscriptions may lead to more question answerings in later periods. In Equation (6), β_{im}^t is log-transformed so that the mean of the log-normal distribution is not restricted by the positive constraints on β_{im}^t . In complement to the temporal dependency captured in previous Equation (5), the correlated Markov model in Equation (6) quantifies within and across activity-type transitions of individual preferences. Together, the model enables one user activity on a latent topic to influence other subsequent user activities in both the short-term and long-term horizons.

3.2. Inference and Orthogonal Topic Regularization

We let $\eta := (W_\theta, \{\Pi^{m'm}\}, \sigma_0)$ denote all the parameters related to the specifications of θ and β , where $W_\theta = (E, W_f, W_i, W_o, b_f, b_i, b_o, W_\mu, W_\sigma)$. Given the above data likelihood and prior specifications, the joint distribution of the data and model parameters is

$$p(y, \theta, \beta, \phi; \eta) = \prod_{i,j,m} \left\{ \prod_t p(y_{ijm}^t | \theta_m^t, \beta_{im}^t, \phi_j) p(\beta_{im}^t | \beta_{i,1:m}^{t-1}; \sigma_0, \Pi) p(\theta_m^t | \theta_m^{1:t-1}; W_\theta) \right\} p(\phi_j). \quad (7)$$

Because of the high-dimensional neural network parameters, we develop a variational expectation-maximization (EM) algorithm (Wainwright and Jordan 2008) to jointly estimate the posterior distribution $p(\theta, \beta, \phi | y; \eta)$ (i.e., the inference step) and optimize the parameters η (i.e., the learning step). The intractable posterior is approximated by a set of variational distributions parameterized by the variational parameters ψ . For the variational distributions, we use the LSTM to model the joint posterior distribution of $\theta^{1:T}$ and the mean-field distributions for β and ϕ ; see Online Appendix A for details. This choice of variational distribution maintains the long-term temporal dependency of θ and achieves high efficiency. The posterior inference is conducted by finding the optimal variational parameter ψ that minimizes the Kullback-Leibler divergence between the variational distribution and the posterior, which is equivalent to maximizing the tractable evidence lower-bound (ELBO) objective (Blei et al. 2017).

With the proposed neural network architecture and the variational distributions, the inference can be conducted efficiently using the re-parametrization trick, which computes the stochastic gradient of an unbiased ELBO estimation through backpropagation (Kingma and Welling 2013). The proposed variational EM algorithm uses the adaptive moment method Adam (Kingma and Ba 2014) to iteratively update the model parameters η and the variational parameters ψ until convergence. This update scheme corresponds to

iteratively performing model fitting and posterior inference. We use the stochastic variational inference (Hoffman et al. 2013, Yin and Zhou 2018) to update the parameters with minibatch data. The minibatch training allows fast parameter updates and efficient memory usage. Online Appendix A contains a complete specification of the modeling and inference structure and the variational EM algorithm.

Our optimization problem may be affected by topic collapse, a common issue observed in neural topic models (Li et al. 2022, Wu et al. 2023). Specifically, we find that the estimated platform-level tag interactions might distribute similarly for two different time steps. This indicates that the algorithm converges to a locally optimal solution. Existing literature addresses the topic collapse issue using orthogonal regularization (Yao et al. 2014). However, this method cannot be directly applied to our problem because it is usually designed for non-dynamic data and aims to reduce the similarity between factor loadings ϕ_k . Thus, we propose a new technique called orthogonal temporal regularization (OTR), which encourages orthogonality and distinctiveness across time t . The OTR involves a hyperparameter $\alpha_m \in [0, 1]$ that controls the regularization strength for the activity m ; we will choose α_m based on the topic quality defined in the following section. The details of the proposed OTR are provided in Online Appendix A.

3.3. Remarks

3.3.1. Incorporating Covariates. In our main specification, we exclude covariates to isolate and evaluate MDNPS's ability to uncover meaningful latent structures and dynamic user preferences solely from observed behavioral patterns. Including external covariates conflates the model's inference power with information driven by external signals, making it harder to assess whether MDNPS can extract semantically coherent topics and realistic preference trajectories purely from sparse, high-dimensional activity data. This baseline approach also reflects many real-world situations where covariates are unavailable, incomplete, or unreliable—ensuring that MDNPS remains applicable in such data-limited settings.

When rich contextual variables are available, MDNPS can be extended to MDNPS-X by directly incorporating platform-level (x^t) or user-level (z_i^t) covariates into the data-generating process. This allows the model to capture both intrinsic behavioral patterns and systematic shifts driven by external factors. We propose to incorporate these covariates into the model as

$$y_{ijm}^t \sim \text{Pois} \left(\sum_{k=1}^K (\theta_{mk}^t + a_{mk}^\top x^t) (\beta_{imk}^t + b_{mk}^\top z_i^t) \phi_{jk} \right) \quad (8)$$

where the parameters $a_{mk} \in \mathbb{R}^{d_x}$ and $b_{mk} \in \mathbb{R}^{d_z}$ capture the effects of platform-level and individual-level

covariates, respectively, on the activity strength for topic k . For example, in our empirical application, we can take x^t as the monthly platform growth metrics and z_i^t as the users' social activities.³

3.3.2. Applicable Domains. The proposed MDNPS is a general-purpose factor model for high-dimensional count data, built on a Poisson likelihood. In our empirical study, we apply it to tokenized textual tags, but the framework can be readily adapted to other types of discrete observations. For instance, it can be used to uncover latent preference for the factors in image data represented by discrete pixel intensities (Zhou et al. 2009) or to model customers' discrete product choices in settings with large assortments and sparse interactions (Gabel and Timoshenko 2022, Liu et al. 2024). Therefore, MDNPS serves as a flexible approach for extracting low-dimensional latent factors from complex, sparse, and high-dimensional count data and uncovering the evolution of population and individual preferences across these factors.

4. Empirical Application

4.1. Research Context

Our application context Zhihu had approximately 100 million monthly active users, 44 million questions, and 2.4 billion answers by the end of 2020 (Ouyang 2021). The key features of Zhihu are similar to those of Quora, a well-known English Q&A platform. Zhihu users can seek knowledge, expertise, experiences, and customized solutions by posting questions. A typical question consists of a title and several tags that indicate the themes of the question. During our study period, tags were created primarily by the experts and experienced users in the early years of Zhihu and remained relatively stable over time; ordinary users without high contributions were generally not permitted to create new tags (Nie et al. 2020). This is also confirmed by the data patterns shown in Figure B.4 of Online Appendix B.

A question usually receives multiple answers ranked by the number of votes they have received. See Figure 2 for an example. Users may subscribe to a newly posted question to actively track the ensuing discussion, receiving notifications whenever new answers appear. Subscribing to a question thus serves as a strong indicator of user consumption interests.⁴ Users can follow any other users, and each user's follower size is a strong indicator of his or her expertise and reputation in this online community.

The Q&As on Zhihu span thousands of different topics, covering all walks of life, including finance, healthcare, the Internet, information technology (IT), education, social issues, and entertainment. According to a 2017 report by iClick and Zhihu,⁵ approximately 70% of users leverage Zhihu for knowledge acquisition

Figure 2. (Color online) Example of a Question and Its Answers

The screenshot shows a Zhihu question titled "如果把库里放到上个世纪 90 年代, 还能不能有巨星的表现?" (If Curry was placed in the 1990s, could there still be superstar performances?). The question has 138 answers. The top three answers are highlighted with red boxes and labeled Answer 1, Answer 2, and Answer 3. Answer 1 is by user 张佳玮 (Zhang Jiawei) and discusses Curry's impact on the league. Answer 2 is by user 演完三点 (Yan Wan San Dian) and discusses the league's state in the 1990s. Answer 3 is by user Air 真心为你 (Air Zhen Xin Wei Ni) and discusses Curry's impact on the league.

and self-improvement, 68% express a desire to subscribe and discuss topics of interest, and 43.1% wish to share their insights and experiences. Therefore, understanding users' topical interests/expertise and facilitating their learning/exploration journey is critical to the platform and user welfare. More generally, given the widespread popularity of Zhihu, its users' behavior and topical trends can provide valuable representations of the broader population, especially among the well-educated young Chinese-speaking demographics.

4.2. Data Description

We have rich data sets on the historical activities of all the registered users from its launch in January 2011

until June 2017. We focus on modeling user answer contributions and question subscriptions. Specifically, answered questions reflect user domain expertise, whereas subscribed questions reflect user consumption preferences.⁶ To ensure sufficiently rich behavioral data for robust and interpretable inference, we construct our sample using the following selection criteria. First, we focus on user activities from the first month to the 30th month since each user's registration. This criterion permits comparing and synchronizing the "personal clocks" across different users. Second, we confine the sample to users who joined the platforms before January 1, 2015, so we have at least 30-month observations per user. Third, for understanding the dynamics in individual users' multifaceted preferences, we include only users with at least one answer contribution and one question subscription activity every three months since joining Zhihu.

These selection criteria resulted in a sample of $N = 14,902$ users, $J = 19,439$ tags, $T = 72$ calendar months, 30-month user lifetime, and $M = 2$ activity types, where we let $m = s$ indicate the content consumption behavior (i.e., subscribing to questions) and $m = a$ indicate the content contribution behavior (i.e., answering questions). We adopt the month as the unit of analysis to balance activity richness and temporal granularity. For the extended MDNPS-X with external covariates, we take the monthly new users, tags, questions, and answers on Zhihu as the platform covariates and the number of monthly new followers and followers of a user as the user covariates.

The resulting sample represents a group of relatively active users on Zhihu, who play a disproportionately important role in driving content creation and sustaining community vitality (DeHaan et al. 2024). By the end of the 30th month on Zhihu, an average sample user contributed 666 answers ($S.D. = 1,121$) and subscribed to 2,075 questions ($S.D. = 4,484$). These questions are related to 19,439 tags, reflecting broad and diverse interests among users. Despite our selection of relatively active users, we still find large activity sparsity within each user and high heterogeneity across users. See Figure B.5 in Online Appendix B for details. These observations confirm the importance of using sophisticated models to understand user preferences effectively in this context.

We use PyTorch as the auto-differentiation tool to compute the stochastic gradient of the ELBO. For each round of parameter update, we use a minibatch of 200 users to compute the objective of the stochastic variational inference. We set three layers of 200 hidden units as our neural network architecture. We apply a small weight decay of 1.2×10^6 on all neural network parameters, following the convention in training deep models.

4.3. Topic Quality and Model Hyperparameters

We use the quality of extracted topics to determine the model hyperparameters (i.e., the number of topics K and the regularization strengths α_a, α_s) because topic quality is critical for deriving meaningful insights from large-scale, high-dimensional data. We compute the *topic coherence* and *topic diversity*, following the existing literature. Topic coherence quantifies the semantic similarity between high-probability words within a topic, indicating how meaningfully they are related to form a cohesive topic (Röder et al. 2015). The coherence of a topic is high if its top tags tend to be associated with the same content. However, topics that collapse into a single topic may have high coherence but low quality. This concern is alleviated by considering topic diversity, which is defined as the proportion of distinct words among the top 25 words across all topics (Dieng et al. 2020). Topic coherence depends on the activity type, whereas diversity does not. Finally, we combine the two metrics by defining the *topic quality* as the product of topic coherence and topic diversity (Dieng et al. 2019, 2020).

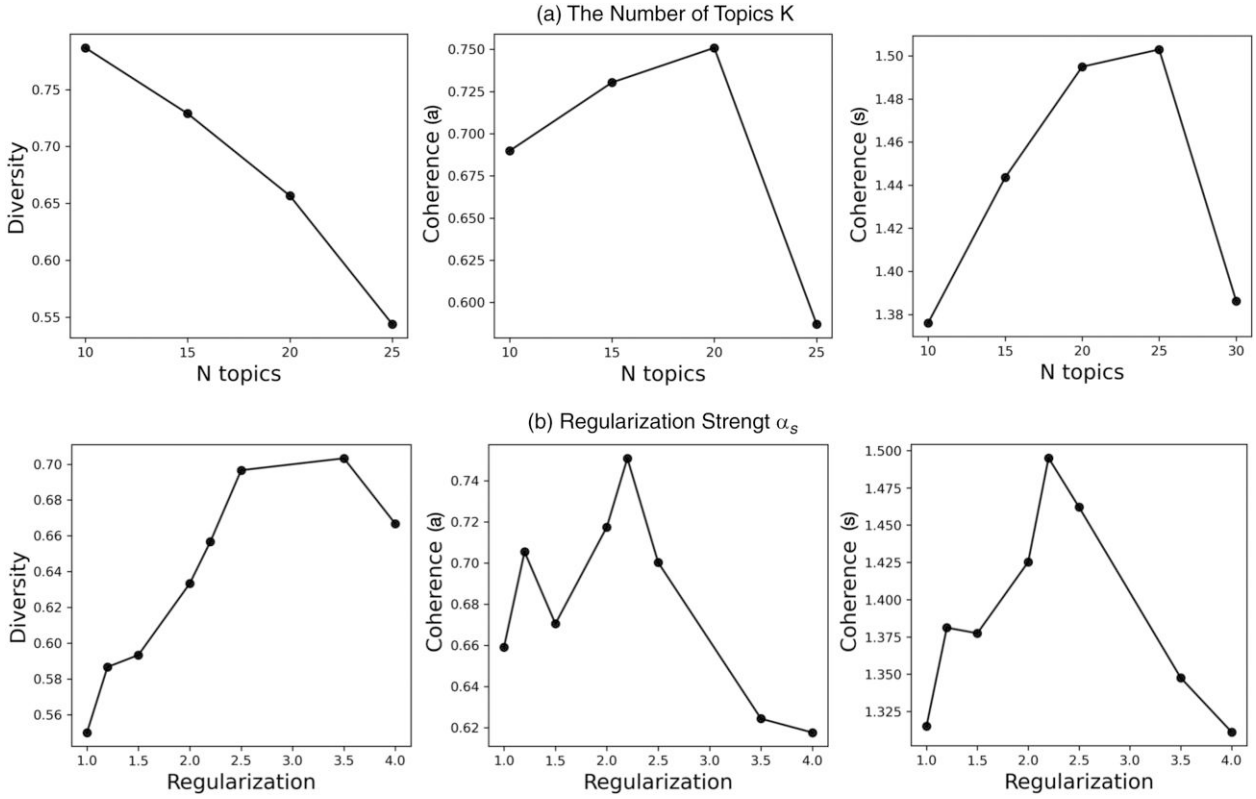
We conduct a grid search over the hyperparameter space $K \in \{10, 15, 20, 25\}$, $\alpha_a \in \{0.05, 0.1, 0.3, 0.5\}$, and $\alpha_s \in \{1.0, 1.5, 2.0, 2.2, 2.5, 3.5, 4.0\}$, using a greedy selection procedure. Specifically, we first determine the optimal value of α_a using only the answering data, because this step does not depend on the choice of α_s . Once α_a is fixed, we proceed to select (α_s, K) using both answering and subscription data. Figure A.2 in Online Appendix A.7 shows the topic coherence and diversity for varying α_a , where $\alpha_a = 0.1$ giving the highest topic coherence, and we then choose α_s and K with $\alpha_a = 0.1$. Figure 3 shows the topic coherence and diversity for varying K with $\alpha_s = 2.2$ and varying α_s with $K = 20$. Overall, in the top panels, topic diversity decreases with K because the high-frequency tags are more likely to appear in different topics when K increases. The coherence peaks around $K = 20$ for both answering and subscribing and then decreases. We also notice in the bottom panels that the topic diversity increases with α_s , which is consistent with our motivation in designing the OTR regularization, whereas the coherence decreases for large α_s because the model is forced to incorporate irrelevant tags into the topics. In summary, among the combinations, we choose $K = 20$, $\alpha_a = 0.1$, and $\alpha_s = 2.2$ because they give the highest topic coherence and most interpretable topics while keeping a relatively high diversity.

5. Model Evaluation

5.1. Benchmark Methods

To understand the performance of the MDNPS in fitting the data set, we consider potential alternative models for benchmark comparison. A naive approach is

Figure 3. Selection of Model Hyperparameters



to model the data of each activity type and time slice with an independent topic model. However, this approach cannot generate consistent topics (as shown in the next subsection) and reduces data efficiency because only a small fraction of users interact with a small subset of tags at each period. Another simplification would be a two-stage model that first learns a topic model and then uses a sequential model to fit the topic dynamics over time and across activities. However, this approach will result in discontinuities and inconsistencies in topic intensity evolution because it does not use information from adjacent time steps or related activities to inform the topic intensities. The advantages of integrating the learning of topics and their preferences evolution into a single, coherent framework have been theoretically and empirically validated in the dynamic topic model literature (Blei and Lafferty 2006).

Given the above discussion, we compare the MDNPS against four benchmark factor models that can be applied to our data context. The first is the Bayesian Poisson Tensor Factorization (BPTF), which was introduced in Schein et al. (2015). The BPTF is a nondynamic factor model that applies to higher-order tensors. It factorizes a tensor in each of its orders by assuming that for each activity type m , the observed count $y_{ijm}^t \sim \text{Pois}(\sum_{k=1}^K \theta_{imk}^{(1)} \theta_{jmk}^{(2)} \theta_{tmk}^{(3)})$. Each latent variable is modeled with an independent prior

distribution, so the BPTF does not capture temporal or cross-activity dependence. The comparison with the BPTF measures the contribution of joint modeling.

The second benchmark is the deep dynamic Poisson factorization model (DDPFA), introduced in Gong and Huang (2017), which is a state-of-the-art dynamic model for count tensors. DDPFA uses deep recurrent neural networks to capture temporal dependence and models the counts of answers or subscriptions by $y_{ijm}^t \sim \text{Pois}(\sum_{k=1}^K \theta_{mk}^t \phi_{jk})$. The DDPFA differs from our MDNPS in two main aspects; first, it models the various user activities independently; second, it infers user preferences only at the aggregated level. The comparison with the DDPFA explores the benefits of modeling the dependency of multiple activities and modeling individual-level preferences.

Finally, we compare with two modified versions of the MDNPS, denoted by the MDNPS\NN and MDNPS\OTR. MDNPS\NN uses the Markov transition, similar to Equation (6), to model the platform dynamics instead of using LSTM. The comparison examines the benefits of incorporating deep learning into the probabilistic model. MDNPS\OTR optimizes the ELBO objective without the OTR regularization. This variation acts as an ablation study to illustrate the effectiveness of our proposed regularization technique in addressing the topic collapse problem.

Note that the topic number K of each benchmark model is also determined using a grid search for $K \in \{10, 15, 20, 25\}$ and the topic quality metrics. BPTF and DDPFA are inferred by the variational inference proposed in the original papers and are optimized by maximizing the ELBOs with stochastic gradient descent until convergence. Sample substantive outputs of benchmark models MDNPS\NN and MDNPS\OTR are available in Online Appendix A.8.

5.2. Topic Quality

We first compare these models on topic coherence, diversity, and quality (as defined in Section 4.3) in Table 1. The MDNPS performs significantly better than the benchmark methods BPTF and DDPFA along all metrics ($p < 0.05$). The much higher coherence of the MDNPS relative to these benchmarks suggests that the MDNPS can better cluster tags with similar semantics, enabling easier and more precise interpretation. For example, the “Relationships” topic has its top words as “Appearance, Intimacy, Childhood, Self-Management, Dating, Height, Goddess,” and the “Sciences & Engineering” topic has its top words as “Agriculture, Civil Engineering, Manufacturing, Computer Major, China Stock Market, Cancer, C/C++.” The much higher diversity of the MDNPS relative to these benchmarks indicates its superiority in flexibly identifying distinctive topics. For example, our extracted topics cover a wide spectrum of Zhihu’s content ecosystems, including E-Commerce, Entertainment, Relationships, Politics, Mobile Device, and Sciences & Engineering. BPTF has the lowest topic quality scores, which indicates the significance of modeling preference dynamics. The improvement of MDNPS over DDPFA is likely due to the cross-activity dependencies in modeling the individual-level dynamics.

Next, we compare MDNPS with two variants to validate our design choices. MDNPS\NN, which models dynamics without neural networks, achieves high diversity but suffers from very low coherence, particularly in question subscriptions, resulting in poor overall quality scores. Conversely, MDNPS\OTR demonstrates high coherence but achieves this by collapsing topics

into a limited set with low diversity, also yielding low-quality scores. Visualizations of the platform-level dynamics θ_{mk}^t estimated by these variants (Online Appendix A.8) reveal distinct patterns that explain these limitations. MDNPS\NN produces non-smooth dynamics with frequent spikes, reflecting its inability to capture long-term trends effectively. Meanwhile, MDNPS\OTR displays highly similar trajectories across multiple topics, which signals topic collapse. These findings confirm that including deep-learning components and OTR regularization significantly improves topic diversity (by addressing the topic collapse problem) and topic coherence (by grouping semantically related words to the same topics).

We also evaluate MDNPS-X that incorporates platform and individual covariates as described in Section 4.2. We find a slight improvement in topic quality: 0.552 for answering questions (coherence is 0.679 and diversity is 0.813) and 1.037 for subscribing questions (coherence is 1.275 and diversity is 0.813). That is, incorporating these covariates in MDNPS can slightly improve its topic diversity but with lower topic coherence. For interested readers, Online Appendix C.7 reports some substantive outputs from MDNPS-X.

5.3. Out-of-Sample Fitting

We finally evaluate these models on fitting out-of-sample user activities from the 31st month of the user lifetime, which was withheld during training of each model.⁷ We use standard metrics for evaluating factor models (Blei et al. 2003, Schein et al. 2019): the log-likelihood of the predictive distribution and the perplexity score (Wallach et al. 2009). Perplexity is computed as the exponentiated and reweighted negative likelihood of the holdout data, defined in the Online Appendix A.6 (i.e., a lower value indicates a better generalization from training data to new data). Thus, both metrics are likelihood based, incorporating point prediction accuracy and uncertainty. Results presented in Table 2 demonstrate that MDNPS significantly outperforms BPTF and DDPFA on all metrics ($p < 0.05$), improving perplexity by more than 50% and increasing log-likelihood by nearly 1.0 for both activity types.

Table 1. Comparison for the Topic Quality Across Models

Metric	BPTF	DDPFA	MDNPS\OTR	MDNPS\NN	MDNPS
Answering questions					
Coherence	0.695	0.474	0.814	0.621	0.751
Diversity	0.237	0.353	0.336	0.710	0.657
Quality	0.164	0.168	0.274	0.441	0.493
Subscribing questions					
Coherence	1.104	1.229	1.118	1.047	1.500
Diversity	0.287	0.460	0.336	0.710	0.657
Quality	0.316	0.565	0.376	0.743	0.985

Note. The bold values indicate the highest and lowest values across the different models.

Table 2. Out-of-Sample Data Fitting Across Models

Metric	BPTF	DDPFA	MDNPS\OTR	MDNPS\NN	MDNPS
Answering questions					
Perplexity	1,892.80	1,535.90	680.90	640.50	579.80
Log-likelihood	−7.87	−7.67	−7.20	−7.05	−6.92
Subscribing questions					
Perplexity	999.60	463.20	232.90	208.70	183.20
Log-likelihood	−7.30	−6.49	−6.20	−5.92	−5.79

Notes. Lower perplexity (higher log-likelihood) indicates a better fit. The bold values indicate the highest and lowest values across the different models.

Therefore, we conclude that the MDNPS can fit and predict the data better than these benchmarks.

6. Substantive Outputs

This section reports the estimated posterior mean of model parameters, including the latent topics ϕ_{jk} , the platform-level topic dynamics θ_{mk}^t , and the user-level preference trajectories β_{mk}^t . We showcase the face validity of MDNPS in reducing complex high-dimensional data into low and interpretable spaces and in providing meaningful substantive insights.

6.1. Topics

Our model depicts the content structure on the platform. The extracted topics summarize the sparse interaction between a large number of users and tags, revealing interpretable content structure at the topic level. We interpret each extracted topic k based on the tags with the highest probability weights in ϕ_k . Figure 4 presents the word clouds of the 20 extracted latent topics using the top 10 tags per topic.⁸ Aligned with the topic quality metrics in the previous subsection, we find that the top tags of each topic are semantically similar and highly coherent. The topics are easy to interpret, except one that comprises a variety of categories, labeled as “Others.” These latent topics loosely correspond to three topical categories. The first category relates to IT and engineering, including “Mobile Device,” “Tech-Finance,” and “Sciences & Engineering.” The second category corresponds to social sciences such as “Culture & Religion,” “History & Novels,” and “Politics.” The third category is related to entertainment and daily life, including “Lifestyle,” “Relationships,” and “Entertainment & Trends.” Overall, these topics represent the information needs, expertise, and leisure interests of the Zhihu user community.

6.2. Platform Topic Trends

We now illustrate how MDNPS enables platforms to understand the collective dynamics of user preferences across topics and activities by tracking $\theta_{mk}^t, m \in \{a, s\}$. The model is well suited to this task because it projects high-dimensional, complex behavioral data into low-dimensional, interpretable representations. By jointly

modeling answer contributions and question subscriptions, we can directly compare θ_{ak}^t and θ_{sk}^t , thereby identifying how content supply and demand evolve and interplay over time.

We compute the normalized monthly popularity of each topic for both content contribution and consumption (i.e., $\theta_{mk}^t / \sum_{t'=1}^T \theta_{mk}^{t'}$), which enables us to track their relative dynamics over time. As shown in the left panel of Figure 5 for the answer contribution, early question answerings (2011–2013), when Zhihu operated by invitation only, centered on IT-related topics such as “Mobile Devices” and “Online Communities,” reflecting a user base of tech-savvy early adopters. After open registration in 2013, topic diversity increased markedly, with rising contributions in areas like “Sciences & Engineering,” “History & Novels,” and “Relationships,” signaling broader public engagement. Popular topics like “Entertainment & Trends” also gradually gained prominence. The right panel of Figure 5 reveals similar shifts in consumption. IT topics dominated early on but declined after 2013.

In Figure 5, some topics display a small but nonzero platform-level intensity during certain periods (e.g., “Relationship” before 2013). A small θ_{mk}^t indicates that the topic at time t is driven mainly by individual niche interests rather than broad platform trends. This sparsity in θ_{mk}^t arises from the sparsity-inducing property of Poisson factor models (Gopalan et al. 2014, Schein et al. 2019), a desirable property for interpretation because it highlights periods when the platform-wide trends are truly evident. Compared with content contribution, consumption patterns display sharper spikes and are more event driven. Importantly, these spikes in consumption tend to lag behind trends in contributions. For example, in “Politics,” contributions gained momentum well before the U.S. presidential election in November 2016, whereas consumption peaked only immediately before the election. Platforms could leverage answer contributions to anticipate emerging trends because content creators and experts are more sensitive to market signals and capable of catalyzing demand.

Further analysis reveals notable mismatches between content supply and demand across topics. To quantify such imbalances, we calculate a topic-specific

Figure 4. (Color online) Word Clouds of the Extracted Topics

ratio Δ_k , defined by the total consumption intensity divided by total contribution intensity as $\Delta_k = \sum_{t=1}^T \theta_{sk}^t / \sum_{t=1}^T \theta_{ak}^t$ for topic k . A higher Δ_k indicates greater unmet demand during this period because the intensity of consumption (θ_{sk}^t) is stronger than that of contribution (θ_{ak}^t). Figure 6 shows $\sum_{t=1}^T \theta_{sk}^t$, $\sum_{t=1}^T \theta_{ak}^t$, and their ratio Δ_k . We observe the largest gaps in “E-commerce,” “Tech-Finance,” “Politics,” and “Culture & Religion.” Such gaps provide platforms with a diagnostic tool for detecting topic-level demand-supply imbalances that are difficult to infer systematically from sparse raw user activities. By interpreting the magnitude of these gaps in light of specific objectives, managers can make more informed decisions. For instance, when the goal is to increase answer response rates, platforms can steer contributor attention toward high-demand, low-supply topics through expert recruitment, targeted incentives, or creator-side recommendations.

6.3. User Preference Dynamics

We now examine the landscape and evolution of individual users’ preferences. The parameter β_{ink}^t provides a clean measure for this purpose because it captures user-specific preference components while netting out platform-level temporal trends. This orthogonalization

enables us to isolate intrinsic shifts in individual interests that would be obscured if relying solely on raw activity volumes.

For ease of interpretation, we develop a unified procedure to track the temporal evolution of β_{ink}^t by embedding users’ topic preferences over time and projecting them into a two-dimensional space. Specifically, for each interest type (i.e., consumption or contribution), we segment each user’s lifespan into five semiannual periods, compute the average topic preference for each period, and apply uniform manifold approximation and projection (McInnes et al. 2018) to project these into a consistent two-dimensional space. Following the approach in Liu and Cong (2023), we include anchor tags, which are Zhihu’s first-level categorization of content that covers the major topical domains on the platform, to contextualize the embeddings. In the projection space, we perform K-means clustering to identify distinct user groups and monitor their transitions over time. This approach enables us to uncover both the structure and dynamics of user interests along a continuous spectrum. More technical details are available in Online Appendix C.2.

6.3.1. Preference Clusters and Migration. Figure 7 shows the resulting consumption and contribution clusters, respectively, which share similar overall

Figure 5. (Color online) Platform-Level Within-Topic Dynamics by Activity Type

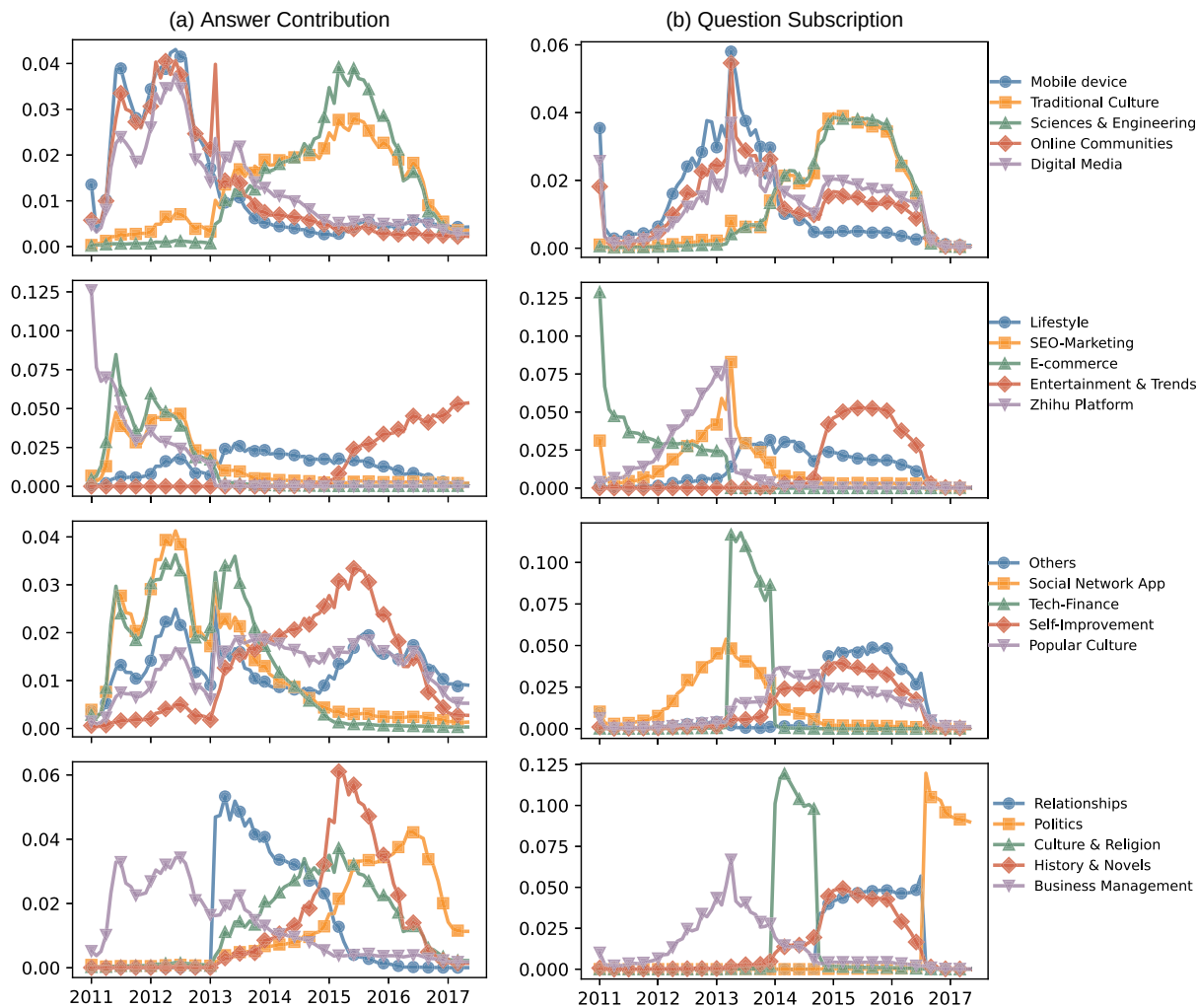


Figure 6. (Color online) Platform-Level Accumulated Topic Strength and Its Ratio

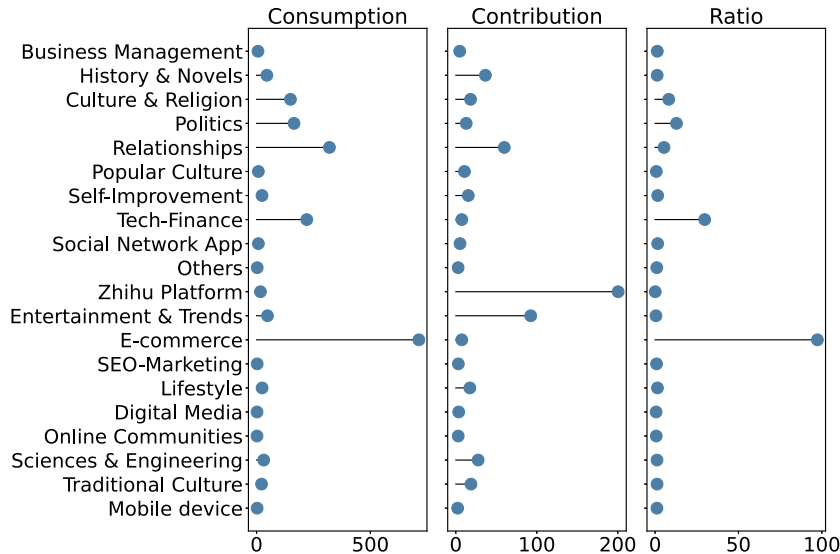
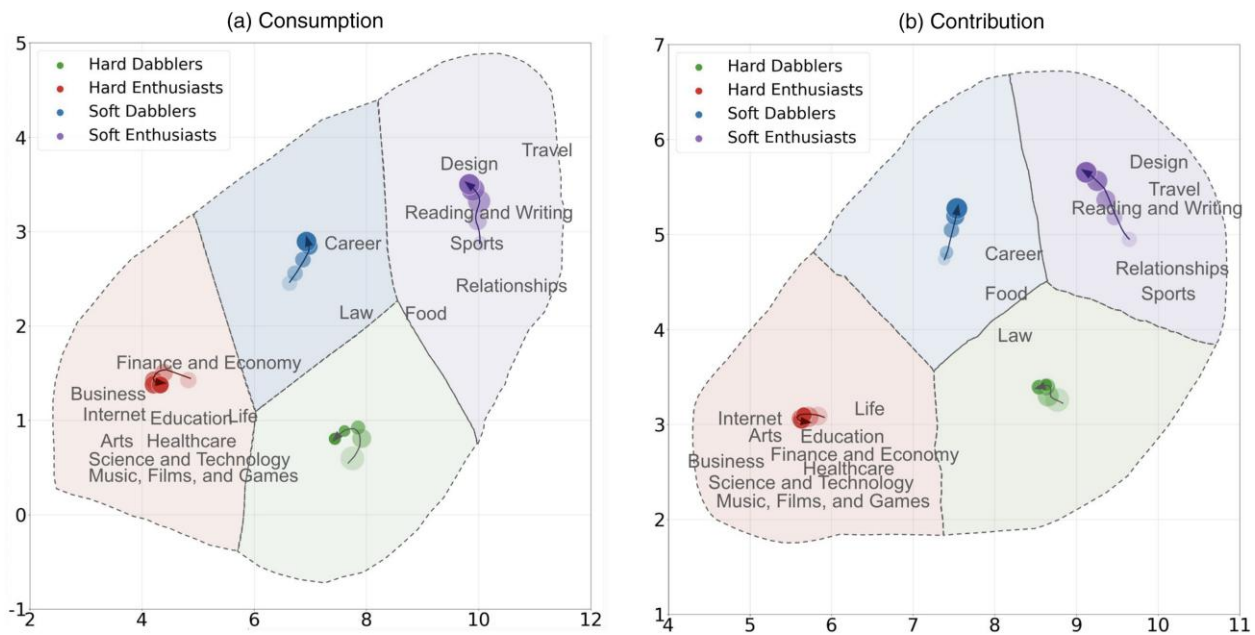


Figure 7. (Color online) Visualizing the Evolution of Individual Preferences

patterns. The dotted lines indicate the boundaries between clusters, and the nodes indicate the centroids of each cluster over time. The anchor tags are represented as text. The entire space can be conceptualized as a spectrum, with hard-topic tags (such as Finance and Economics, Science and Technology, and Healthcare) concentrated in the lower-left fringe, whereas soft-topic tags (such as Travel, Design, and Sports) are in the upper-right corner. We name the lower-left and upper-right clusters as *Hard Enthusiasts* (HE) and *Soft Enthusiasts* (SE) based on the proximity and alignment of individual preference embeddings and the anchor tag embeddings. The two central clusters exhibit ambiguous and mixed interests. We name them as *Hard Dabblers* (HD) and *Soft Dabblers* (SD), according to their inclinations toward hard and soft topics over periods.

Figure 7 also suggests moderate migration within each cluster. To further investigate, Table 3 reports the cluster proportions across the first, third, and fifth semiannual periods. In consumption, only 35% of users were initially classified as *Soft Dabblers* or *Soft Enthusiasts*, but this share increased to 65% by the fifth period, indicating a general shift toward soft topics. The notable exception is *Hard Enthusiasts*, whose proportion remains stable at around 22%. For content contribution, however, even *Hard Enthusiasts* lose substantial ground to soft-topic clusters, declining from 43% in the first period to 13% by the fifth.

We next examine how individual users' cluster affiliations evolve over time. The trajectories are visualized in Figures C.6 and C.7 in Online Appendix C.3,

and we summarize the main findings here. In consumption, users who begin as *Hard Enthusiasts* show relatively strong loyalty, with 22.1% remaining in the same cluster through the fifth period, reflecting sustained interests in hard topics. In contrast, *Dabblers*—those with initially ambiguous interests—are more fluid, often transitioning toward *Soft Enthusiasts*. A slightly different pattern emerges in content contribution; only 15.1% of *Hard Enthusiasts* remain by the fifth period, indicating that sustained contributions to hard topics are even rarer than continued consumption in these topics.⁹ Taken together, these results consistently highlight two managerial implications. First, users—particularly *Dabblers*—tend to gravitate toward soft topics, suggesting that platforms may benefit from exposing broad, accessible content to users without strong initial interests. Second, although hard-topic enthusiasts often remain loyal consumers in these topics, sustaining their contributions proves more challenging. For platforms, this indicates a need to offer targeted incentives to sustain expert engagement in complex domains over time.

6.3.2. Consumption-Contribution Discrepancy. Our findings thus far reveal nuanced correlations and distinctions between users' consumption and contribution interests, demonstrating the importance of examining the two types of activities jointly. To advance this understanding, we now consider individual users' preferences for consumption and contribution simultaneously. Based on the four user clusters, there are 16 possible consumption(S)-contribution(A) combinations.

Table 3. Cluster Size Over Periods

Clusters	Consumption			Contribution		
	1st period	3rd period	5th period	1st period	3rd period	5th period
Hard enthusiasts	0.210	0.253	0.232	0.430	0.241	0.129
Hard dabblers	0.438	0.161	0.120	0.268	0.290	0.193
Soft dabblers	0.183	0.191	0.308	0.121	0.187	0.340
Soft enthusiasts	0.168	0.395	0.340	0.181	0.281	0.340

Figure 8 shows the distribution of these combinations in users’ first and last semiannual periods. Diagonal elements represent users with aligned consumption and contribution preferences, whereas upper- and lower-diagonal elements reflect users whose contributions are more or less hard-oriented, respectively, than their consumption. We observe that among the users contributing to hard topics (*Hard Enthusiasts (A)*), those consuming soft topics (*Soft Enthusiasts(S)*, *Soft Dabblers(S)*) more than doubled. This suggests that soft topics can be provided for subscription to the users who primarily answer hard topics. Another clear migration pattern is toward soft interests. The most dominant combination changed from *Hard Dabblers* in both dimensions (22.9%) to *Soft Enthusiasts* in both dimensions (17.98%).

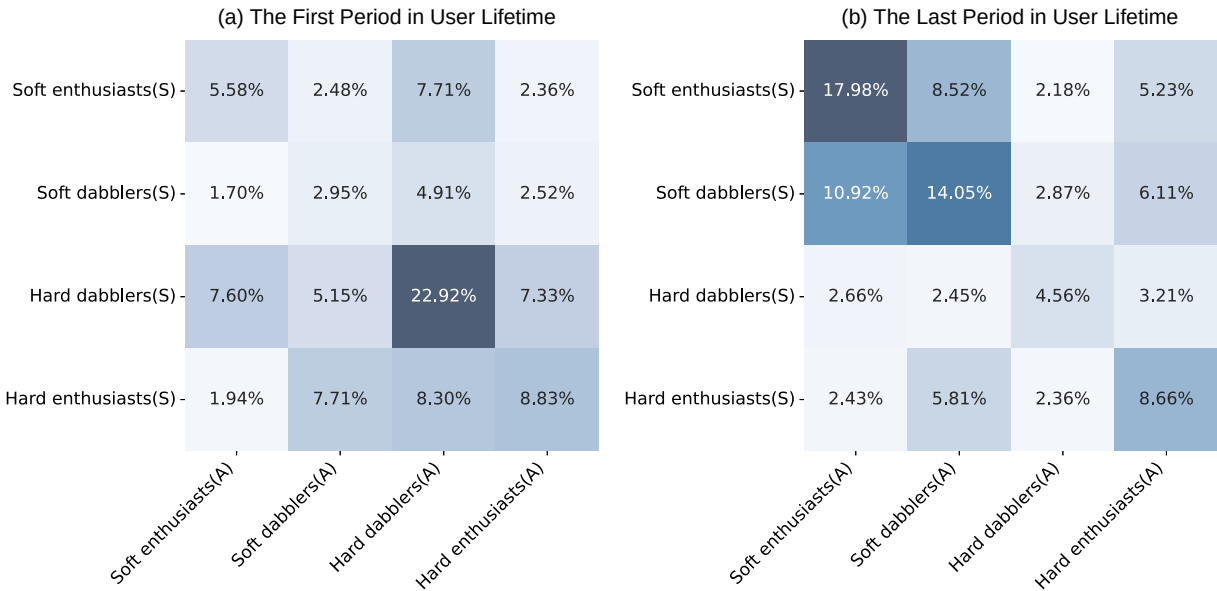
We further compute an L_2 norm-based (continuous) discrepancy metric across user lifetime; $\|\beta_{ia}^t - \beta_{is}^t\|_2$. Figure C.11 in Online Appendix C.5 reports the average discrepancy across user segments defined by the final-period cluster membership, revealing consistent heterogeneity. Here, we summarize the main implications from the data patterns. Users drawn to hard topics tend to develop increasing divergence between

what they consume and contribute, often contributing hard content while consuming softer material. In contrast, soft-topic users show stronger alignment over time. These results suggest the need for activity-dependent content promotions.

6.3.3. The Decomposition of Individual Activities. MDNPS enables platforms to understand to what extent individual users’ activities are associated with their preferences (i.e., $\beta_{imk}^{t_i}$) relative to the aggregated platform trends (i.e., $\theta_{mk}^{t_0+t_i}$). Such insights can help to understand whether a user’s behavior is preference or trend driven. We approximate this relationship using a ratio: $R_{imk}^{t_i} = \beta_{imk}^{t_i} / \theta_{mk}^{t_0+t_i}$ for $m \in \{a, s\}$, which is high when individual interest dominates the platform trend. In the following, we show some findings that confirm the face validity of our model outputs.

For ease of illustration, we compute the average $R_{imk}^{t_i}$ across time for each user-topic-activity combination, denoted by $\bar{R}_{imk}$. We find that $\bar{R}_{imk}$ exceeds 1 in more than 90% of observations, indicating that user activities are more strongly linked to individual preferences

Figure 8. (Color online) The Proportions of Different Consumption-Contribution Subgroups



than to aggregate trends. In addition, the average $\bar{R}_{imk}$ is significantly higher for answer contribution than for question subscription ($p < 0.01$), suggesting that, intuitively, consumption aligns more closely with platform trends than contribution. This suggests that platforms can adopt trend-driven nudges to encourage subscription while personalizing content offerings more strongly for answering.

We further explore the heterogeneity across topics. Figure 9 displays the average contribution-based ratio across users over their lifetime for eight sample topics. There are notable increasing trends for several topics, such as “Online Communities” and “Social Network App,” indicating sustained and inherent user interests in IT-related areas. This aligns with the fact that many active users in our study window are IT professionals. In contrast, general-interest topics, such as “Politics” and “Culture & Religion,” show declining trends. This indicates that, intuitively, user interests in general-interest topics are more likely to be associated with platform trends than personal preferences, especially in the later stages of the user’s lifetime. Platforms can adjust the weights of population-level and individual-level factors in deciding content offerings, depending on the identified heterogeneity across time and topics.

6.4. User Preferences and Follower Sizes

These estimated individual preferences allow us to explore how user preferences are associated with managerially relevant individual outcomes. Here, we consider each user’s follower size at the end of their lifetime, denoted by $Follower_i$. Follower size is an important indicator of user success on digital content platforms and is frequently utilized by platforms to determine various decisions, such as identifying content experts and promoting users (Tian et al. 2024).

We first regress the log-transformed $Follower_i$ on user i ’s contribution- and consumption-based cluster

affiliations observed at the last semiannual period. Table 4 presents the estimation results while setting the *Soft Enthusiasts* cluster as the baseline group. All model coefficients are significantly positive ($p < 0.05$), with *Hard Enthusiasts* showing the highest follower size, followed by *Hard Dabblers*, *Soft Dabblers*, and *Soft Enthusiasts*. This suggests that users with hardcore expertise/interests tend to attract more followers than users with soft-oriented preferences. Intuitively, we find that contribution preferences are more strongly related to user follower size than consumption preferences based on the adjusted R-square values and the magnitudes of the coefficients. We also replicated this analysis using user cluster affiliations at the first and third semiannual periods. The results are reported in Online Appendix C.6. The patterns remain qualitatively consistent, although the effects of early-stage preferences on follower size are weaker. This finding confirms that users need time to explore and form their preferences/expertise on the platform.

Next, we quantify the relationship between follower size and individual preferences (β_{imk}^{ti}), the ratio metric ($\bar{R}_{imk}$) defined previously. We specify the log-transformed $Follower_i$ as

$$\ln(Follower_i + 1) = \alpha + \sum_{m \in \{s, a\}} \sum_{k=1}^K (\gamma_{mk} \bar{R}_{imk} + \eta_{imk} \bar{\beta}_{imk}) + \epsilon_i, \quad (9)$$

where $\bar{\beta}_{imk}$ denotes the average β_{imk}^{ti} across times for each user-topic-activity combination. This model contains 80 independent variables that capture user consumption and contribution interest across topics. The model fits the data reasonably well, with an R-square of 0.344. The point estimates are shown in Table 5, in which the first (last) two columns are variables related to content contribution (consumption). Notably, the coefficients of contribution $\bar{\beta}_{iak}$ are uniformly positive,

Figure 9. (Color online) The Evolution of Individual-Platform Activity Ratio for Samples Topics

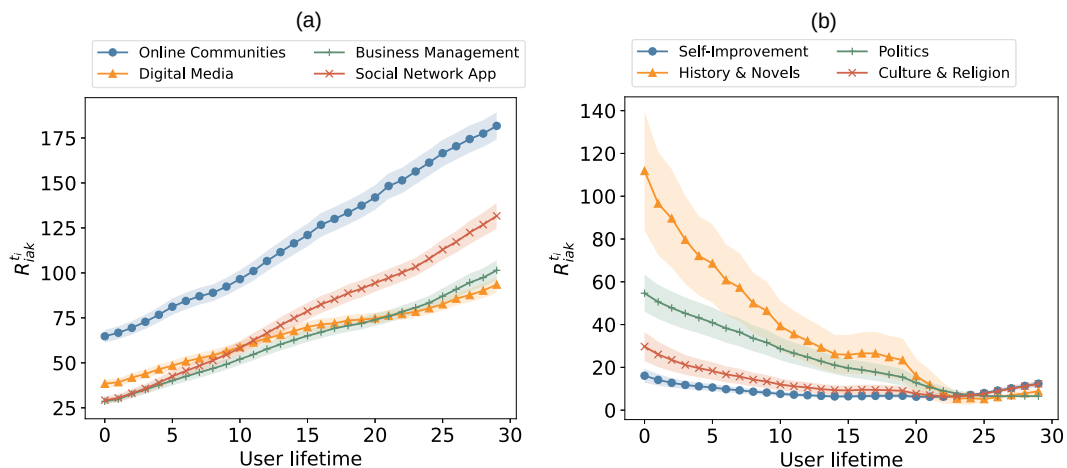


Table 4. Regression of Follower Size on User Eventual Cluster Affiliations

Dep. Var.	$\ln(\text{Follower} + 1)$		
	(1)	(2)	(3)
Intercept	3.172*** (0.036)	3.317*** (0.037)	3.007*** (0.042)
SoftDabblers ^A	0.343*** (0.051)		0.295*** (0.052)
HardDabblers ^A	1.256*** (0.069)		1.021*** (0.073)
HardEnthusiasts ^A	2.114*** (0.060)		1.889*** (0.064)
SoftDabblers ^S		0.379*** (0.054)	0.117** (0.054)
HardDabblers ^S		1.154*** (0.073)	0.632*** (0.075)
HardEnthusiasts ^S		1.230*** (0.058)	0.616*** (0.060)
Adj. R^2	0.087	0.036	0.096

Notes. The table reports regressions of log-transformed follower size on user contribution- and consumption-based cluster affiliation at the last semiannual period. The follower size is the cumulative follower count of each user by the end of their 30th month on the platform.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

with 11 of them being statistically significant ($p < 0.05$), whereas the majority of the coefficients of $\bar{R}_{iak}$ are negative. These findings suggest that although active content contribution can increase user popularity, diverging significantly from the prevailing trends on

Table 5. Relationship Between R_{imk} and Follower Size

Topics	Estimated coefficients			
	$\bar{R}_{iak}$	$\bar{\beta}_{iak}$	$\bar{R}_{isk}$	$\bar{\beta}_{isk}$
Mobile device	-0.007*	0.782**	-0.002	0.350
Traditional culture	-0.001	0.601***	0.001	-0.164
Sciences and engineering	0.011	0.362**	0.002	0.163
Online communities	-0.002	0.182	-0.000	-0.072
Digital media	-0.002	0.326	0.001*	-0.115
Lifestyle	0.035	0.248	-0.001	0.268*
SEO-marketing	-0.007*	0.783**	0.001	-0.210
E-commerce	-0.001	0.130	-0.000**	0.314*
Entertainment and trends	-0.004***	0.366***	-0.001	0.152
Zhihu platform	0.001	0.012	-0.001	0.112
Social network app	-0.001	0.277	-0.001	-0.201
Tech-finance	-0.003**	0.484**	-0.000	0.270
Self-improvement	0.037**	0.246	-0.001	0.239
Popular culture	0.045	0.102	0.004	-0.067
Relationships	-0.000	0.955***	-0.000***	-0.420***
Politics	0.004	0.413**	0.000	-0.112
Culture and religion	0.002	0.585***	0.000	0.097
History and novels	0.003*	0.572***	-0.000	0.162
Business management	-0.018***	0.884***	-0.000	0.316**
Others	-0.007	0.355	-0.001*	-0.124

Note. The table reports the point estimates of a regression model of log-transformed Follower_i on 80 preference-related variables, including the average individual preferences $\bar{\beta}_{iak}$ and the ratio metric $\bar{R}_{imk}$ defined previously for 20 topics and two activity types (i.e., consumption (s) and contribution (a)).

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

the platform may not yield the desired outcomes. Creators with specialized and niche expertise should mindfully design the content and style of their contributions according to the general trends on the platform. As expected, the coefficients of consumption-related variables are less significant and have smaller magnitudes than those of contribution-related variables. This further confirms that the contribution preferences are more influential to user success than the consumption preferences.

6.5. Interpretable Embeddings and Content Recommendation

We have so far illustrated that MDNPS yields transparent, compact, and semantically coherent summaries of user behavior. Such representations can also be incorporated as structured inputs into predictive black-box models and language models for downstream tasks. This subsection uses content recommendation as a simple illustration. Recall that MDNPS can disentangle the embeddings of content and users at the platform and individual levels. Thus, recommendations can be tilted toward individual tastes, broader trends, or specific activities such as answering versus subscribing.

We illustrate this application by predicting questions a user will interact with (answer or subscribe to) in month $t + 1$. Each question d on Zhihu can be represented as a topic vector $\psi_d = (\psi_{d1}, \psi_{d2}, \dots, \psi_{dK})$, with $\psi_{dk} = \sum_{j \in \mathcal{J}_d} \phi_{jk}$ summing over tags $\mathcal{J}_d$ of the question, and each user i 's activity-specific preference trajectory is captured by β_{im}^t . We define the predicted utility user i derived from interacting with question d as $u_{imd}^{t+1} = \sum_{k=1}^K \theta_{mk}^{t+1} \beta_{imk}^{t+1} \psi_{dk}$ (Aggregate) or $u_{imd}^{t+1} = \sum_{k=1}^K \beta_{imk}^{t+1} \psi_{dk}$ (β -only). The Aggregate approach incorporates both platform trends and individual tastes, and the β -only approach uses only the preferences not captured in the overall trend. It enables adjusting the influence of population and individual preferences in recommendations and offering content toward specific user activity.

To approximate recommendation quality, we compare the semantic similarity between the top five recommended questions and those that users actually interacted with (subscribed to or answered) in the month $t + 1$. High similarity indicates greater alignment between the recommended content and the user's subsequent engagement. Because we observe only subscription and answer behaviors instead of exposure or browsing, this metric should be interpreted as an alignment with future observable engagement, not the user's full underlying interests. The metric is measured by computing the cosine similarity of the question text's semantic embeddings, which is obtained by encoding the text with the distiluse-base-multilingual-cased-v2 model from Sentence-Transformers (Reimers and Gurevych 2019), a pretrained

Table 6. Cosine Similarity of Recommended and Actual Interacted Questions

Activity	Aggregate	β -only	Random
Answering	0.575 (0.017)	0.568 (0.014)	0.518 (0.015)
Subscription	0.574 (0.016)	0.520 (0.009)	0.504 (0.010)

Notes. The table reports the average cosine similarity across 100 randomly selected users under each recommendation method, and the standard deviations are in parentheses. Bold text signifies key findings that are discussed in Section 6.5.

language model. For each question recommending to user i , we define its similarity as the highest cosine similarity with the set of questions the user interacted with at time $t + 1$. We then take the highest similarity across all five recommended questions for that user. This procedure is repeated for 100 randomly sampled users.

Performance comparisons of the Aggregate and the β -only approaches against a baseline random selection are provided in Table 6. Relying only on tag-related activity data, both the Aggregate and β -only versions of our model outperform the baseline ($p < 0.01$). The cosine similarity of the baseline random selection remains relatively high because the embeddings primarily capture the similar format of Zhihu questions (e.g., length, word choice, question style), whereas the incremental similarity measure better reflects topic similarity. For answering, the β -only approach performs nearly as well as the Aggregate model ($p > 0.1$), consistent with the idea that contribution is driven primarily by individual interests. For subscriptions, platform trends play a larger role, and the Aggregate model performs significantly better than the β -only version ($p < 0.01$). These results highlight that our interpretable embeddings, even without rich textual features, can assist activity-specific, transparent recommendations. As a further illustration, Table 7 lists the top recommended questions by the Aggregate approach for an example user to subscribe. Despite

different question tags, the recommended questions are largely aligned with the questions actually subscribed to by this user at the topic level.

We acknowledge that this illustration focuses on predicting answering and subscription behaviors and does not extend to broader user activities, such as reading or browsing. Within this focused scope, the evidence suggests that transparent embeddings can add flexibility and interpretability to content suggestions using sparse tag-related activity data. In Online Appendix C.8, we conduct an additional analysis and show that incorporating MDNPS embeddings as augmented inputs to black-box predictive models improves prediction accuracy. This result suggests that our embeddings encode complementary information from the raw data that is not automatically captured by standard black-box models.

7. Discussion and Conclusion

This paper introduces MDNPS to enhance the understanding of user preferences on digital platforms. Methodologically, MDNPS advances the modeling of high-dimensional, sparse, and heterogeneous user interaction data by combining Bayesian principles with deep learning in a dynamic Poisson factor framework. It captures time-varying, activity-specific, and hierarchical user activities, yielding a compact and interpretable representation of user preferences. These latent features could support data mining, downstream decision-making, and applications that require meaningful behavioral signals but cannot process raw interaction histories. Our proposed variational Bayes algorithm enables scalable inference of both neural network parameters and latent structures, making the model suitable for large-scale, sparse, and high-dimensional datasets.

We conduct a comprehensive empirical application to Zhihu data. MDNPS significantly improves in-sample and out-of-sample data fitting compared

Table 7. Recommended and Subscribed Questions for a Sample User

Recommended	Subscribed
• Is Bitcoin worth investing in? Why? (tags: Virtual Currency, Currency, Bitcoin) • What books on media theory and the writing styles of famous journalists are worth reading for someone entering the media industry? (tags: Communication, Journalism, and Communication Master's, Journalism Practice, Journalism Industry) • What do the spouses of IT tycoons do? (tags: Jack Ma, Kai-Fu Lee, William Ding) • What travel routes are recommended in Europe for art and painting enthusiasts? (tags: Artwork, Painting Appreciation, Modern Art, European Culture) • How large is the talent gap in the domestic medical device R&D sector? (tags: R&D, Talent, Computer Technology, Technical Talent)	• Why is Bitcoin so popular? Is it reliable? Can you explain how modern currency is created from a financial and economic perspective? (tags: Currency, Bitcoin) • What graduate programs are suitable for Internet product managers looking to further their education? (tags: University Major Selection, IT Companies) • How do you persuade a visual designer to revise a design they are very satisfied with? (tags: Teamwork, Workplace Communication) • Which IT companies in Guangzhou are engineer-driven? (tags: IT Companies)

with existing benchmark models. Beyond predictive accuracy, MDNPS effectively summarizes users' sparse interactions with topic tags into semantically coherent topics that reflect the platform's topical spectrum. At the platform level, the model reveals meaningful interest dynamics that closely synchronize major external events or societal trends. At the individual level, MDNPS uncovers distinctive preference trajectories and user segments, shedding light on how early interests and expertise shape long-term behaviors and user success on the platform. For instance, we find that users contributing to hard topics tend to attract more followers. Although intrinsic preferences primarily drive user engagement, alignment with broader platform trends, particularly in content creation, also plays a critical role in amplifying user success. These insights would be difficult to obtain without MDNPS's capacity to model high-dimensional, complex, and multifaceted user behavioral data.

Our demonstration of using MDNPS outputs for interpretable content recommendation illustrates how the model's embeddings can be applied in practice. Even when relying only on sparse tag-level inputs, the model delivers substantial improvements in predictive alignment and offers transparent, activity-specific recommendation levers. This exercise is not meant as a head-to-head comparison with state-of-the-art black-box recommenders but rather as a proof-of-concept that the interpretable latent features uncovered by MDNPS can serve as both actionable managerial tools and valuable inputs for downstream applications. This bridge between methodological innovation and practical utility illustrates the broader potential of MDNPS for advancing both research and platform decision-making.

Our study has several limitations and opens promising avenues for future research. First, on the methodological front, the proposed main model is unsupervised, although we incorporate platform and user features in a model extension. Future work can enhance the modeling of user preferences by incorporating richer contextual information. When the content-level attributes for topic k (e.g., its entertainment and informativeness) are available, they can be incorporated in modeling the Poisson rate of MDNPS. Furthermore, the current model captures the dependency among users indirectly through a hierarchical model. When users' social network information is available, their dependencies can be modeled directly, for example, by updating a user's parameter β_{imk}^t through aggregation of their neighbors' parameters, such as using the mean pooling from the graph convolutional network (Kipf and Welling 2016).

Second, the estimated preference parameters are conditional on the platform's exposure mechanism. Hence, they conflate intrinsic preferences with algorithmic exposure from personalized recommendations and reflect the users' *revealed* preferences. Although

this limitation may be less consequential given the descriptive nature and methodological focus of our study, future research could explicitly model the exposure process and control for potential confounders. The proposed MDNPS-X framework is designed to facilitate such applications. Another interesting direction to enhance causal understanding is to integrate the latent factors identified by MDNPS into structural or panel-based models to estimate the causal effects (Ruiz et al. 2020, Puranam et al. 2021). For example, latent individual preferences could inform the utility specification in structural demand models, enabling counterfactual evaluations of content recommendation strategies.

Third, our current framework assumes that the topic-tag distributions (ϕ_k) are time-invariant and attributes temporal variation in activity counts primarily to evolving preferences ($\theta_{mk}^t, \beta_{imk}^t$). Traditional dynamic topic models, in contrast, allow the topics themselves to evolve over time (Blei and Lafferty 2006). A promising future direction is to explicitly capture topic evolution by modeling ϕ_k^t as time-varying, for example, by using deep neural networks to parameterize its dynamics in a manner analogous to our modeling of platform trends. This extension could reveal how the composition and relative importance of tags shift within each topic over time and help platforms proactively refresh content offerings related to specific topics.

Finally, on generalizability, our study focuses on users with relatively frequent content interactions, so the findings may not extend to less active consumers. Future research could assess the model's robustness with alternative groups or extend it across multiple platforms to account for preference heterogeneity. Moreover, as richer data becomes available, future research could expand beyond contributions and subscriptions to incorporate additional engagement metrics such as viewing, voting, and sharing.)

Acknowledgments

The authors thank Yixing Chen, John Lalor, Hui Li, and Khaled Boughanmi for thoughtful comments and the seminar participants at the 2024 China India Insights Program Conference and the ISMS Marketing Science conference for their feedback. The authors thank the senior editor, associate editor, and anonymous reviewers for constructive suggestions in the review process.

Endnotes

¹ Traditional Bayesian methods, despite their transparent model structures and rigorous inference, often face a rigidity issue because of prior and likelihood specification for complex data (Dew et al. 2024). In comparison, deep neural networks excel at fitting complex nonlinear patterns but often fail in interpretability because of their black-box structure (Goodfellow et al. 2016).

² In practice, the LSTM in Equation (4) can be stacked to form an L -layer deep LSTM as $(h_1, h_2, \dots, h_T) = f_L f_{L-1} \dots f_1(\theta^0, \theta^1, \dots, \theta^{T-1})$ to improve flexibility. We adopt $L = 3$ in our empirical study.

³ We do not include covariates at the item (j) level here to balance flexibility with interpretability. This ensures that the loading ϕ_{jk} directly reflects an item's importance in topic k . Incorporating item covariates would mix the intrinsic item importance with the features' predictive power, making the inferred topics less interpretable.

⁴ Although users can also subscribe to tags, such behavior occurs infrequently and offers limited information about evolving preferences. In practice, most users subscribe to a large set of tags at the time of joining Zhihu. Because the overall tag set has remained largely stable and has not expanded substantially over time, the incidence of subsequent subscriptions is low.

⁵ The original report (in Chinese) is currently unavailable online. However, some of its key findings and statistics (in English) can be found here: <https://www.dragonsocial.net/blog/zhihu-marketing-guide/>.

⁶ We do not model the activity of question contribution because of data sparsity (an average sample user contributed only 1.311 questions in an average month).

⁷ For MDNPS, we compute the preference parameters for the 31st month using the transition dynamics in Equations (5) and (6).

⁸ The list of top 10 tags per topic is shown in Table C.1 of Online Appendix C.1.

⁹ For interested readers, we visualize the estimated four transition matrices of the correlated HMM that capture the evolution of individual preferences. Overall, we find similar migration patterns between hard and soft topics. Details are presented in Online Appendix C.

References

- Atchison J, Shen SM (1980) Logistic-normal distributions: Some properties and uses. *Biometrika* 67(2):261–272.
- Blei DM, Lafferty JD (2006) Dynamic topic models. Cohen W, Moore A, eds. *Proc. 23rd Internat. Conf. Machine Learn.* (Association for Computing Machinery, New York), 113–120.
- Blei DM, Kucukelbir A, McAuliffe JD (2017) Variational inference: A review for statisticians. *J. Amer. Stat. Assoc.* 112(518):859–877.
- Blei DM, Ng AY, Jordan MI (2003) Latent dirichlet allocation. *J. Machine Learn. Res.* 3(Jan):993–1022.
- Cain P (2022) Modelling short-and long-term marketing effects in the consumer purchase journey. *Internat. J. Res. Marketing* 39(1):96–116.
- Chakraborty I, Kim M, Sudhir K (2022) Attribute sentiment scoring with online text reviews: Accounting for language structure and missing attributes. *J. Marketing Res.* 59(3):600–622.
- Cheng Z, Lee D, Tambe P (2022) InnoVAE: Generative AI for understanding patents and innovation. Preprint, submitted April 13, <http://dx.doi.org/10.2139/ssrn.3868599>.
- DeHaan E, Kim J, Lourie B, Zhu C (2024) Buy now pay (pain?) later. *Management Sci.* 70(8):5586–5598.
- Dew R, Ansari A, Li Y (2020) Modeling dynamic heterogeneity using gaussian processes. *J. Marketing Res.* 57(1):55–77.
- Dew R, Padilla N, Luo LE, Oblander S, Ansari A, Boughanmi K, Braun M, et al. (2024) Probabilistic machine learning: New frontiers for modeling consumers and their choices. Preprint, submitted April 16, <http://dx.doi.org/10.2139/ssrn.4790799>.
- Dhillon PS, Aral S (2021) Modeling dynamic user interests: A neural matrix factorization approach. *Marketing Sci.* 40(6):1059–1080.
- Dieng AB, Ruiz FJ, Blei DM (2019) The dynamic embedded topic model. Preprint, submitted July 12, <https://arxiv.org/abs/1907.05545>.
- Dieng AB, Ruiz FJ, Blei DM (2020) Topic modeling in embedding spaces. *Trans. Assoc. Comput. Linguist.* 8:439–453.
- Duvvuri SD, Ansari A, Gupta S (2007) Consumers' price sensitivities across complementary categories. *Management Sci.* 53(12):1933–1945.
- Fox V (2007) Pew/internet study finds most americans get their answers from the internet. <https://searchengineland.com/pewinternet-study-finds-most-americans-get-their-answers-from-the-internet-13028>.
- Gabel S, Timoshenko A (2022) Product choice with large assortments: A scalable deep-learning model. *Management Sci.* 68(3):1808–1827.
- Goli A, Chintagunta PK, Sriram S (2022) Effects of payment on user engagement in online courses. *J. Marketing Res.* 59(1):11–34.
- Gong C, Huang W-B (2017) Deep dynamic poisson factorization model. *Advances in Neural Information Processing Systems* 30.
- Goodfellow I, Bengio Y, Courville A (2016) *Deep Learning* (MIT Press, Cambridge, MA).
- Gopalan PK, Charlin L, Blei D (2014) Content-based recommendations with poisson factorization. Ghahramani Z, Welling M, Cortes C, Lawrence ND, Weinberger KQ, eds. *Advances in Neural Information Processing Systems*, vol. 27 (Neural Information Processing Systems Foundation, Red Hook, NY).
- Guadagni PM, Little JD (1983) A logit model of brand choice calibrated on scanner data. *Marketing Sci.* 2(3):203–238.
- Guo D, Chen B, Zhang H, Zhou M (2018) Deep poisson gamma dynamical systems. Bengio S, Wallach HM, Larochelle H, Grauman K, Cesa-Bianchi N, Garnett R, eds. *Advances in Neural Information Processing Systems*, vol. 31 (Neural Information Processing Systems Foundation, Red Hook, NY).
- Hemann C, Burbary K (2013) *Digital Marketing Analytics: Making Sense of Consumer Data in a Digital World* (Pearson Education, Indianapolis).
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Comput.* 9(8):1735–1780.
- Hoffman MD, Blei DM, Wang C, Paisley J (2013) Stochastic variational inference. *J. Machine Learn. Res.* 14(1):1303–1347.
- Horrigan JB (2001) Online communities. PEW RESEARCH CENTER.
- Jacobs B, Fok D, Donkers B (2021) Understanding large-scale dynamic purchase behavior. *Marketing Sci.* 40(5):844–870.
- Kingma DP, Ba J (2014) Adam: A method for stochastic optimization. Preprint, submitted December 22, <https://arxiv.org/abs/1412.6980>.
- Kingma DP, Welling M (2013) Auto-encoding variational bayes. *ICLR*. <https://arxiv.org/abs/1312.6114>.
- Kipf TN, Welling M (2016) Semi-supervised classification with graph convolutional networks. Preprint, submitted September 9, <https://arxiv.org/abs/1609.02907>.
- Koksal I (2020). The rise of online learning. Accessed July 27, 2024, <https://www.forbes.com/sites/ilkerkoksal/2020/05/02/the-rise-of-online-learning/?sh=6c9a2daa72f3>.
- Li H, Ma L (2020) Charting the path to purchase using topic models. *J. Marketing Res.* 57(6):1019–1036.
- Li Y, Wang C, Duan Z, Wang D, Chen B, An B, Zhou M (2022) Alleviating “posterior collapse” in deep topic models via policy gradient. *Adv. Neural Inform. Processing Systems*, vol. 35 (Curran Associates, Inc., Red Hook, NY), 22562–22575.
- Liberali G, Ferecatu A (2022) Morphing for consumer dynamics: Bandits meet hidden markov models. *Marketing Sci.* 41(4):769–794.
- Liu X (2023) Deep learning in marketing: A review and research agenda. *Artificial Intelligence Marketing* 20:239–271.
- Liu J, Cong Z (2023) The daily me versus the daily others: How do recommendation algorithms change user interests? Evidence from a knowledge-sharing platform. *J. Marketing Res.* 60(4):767–791.
- Liu L, Dzyabura D (2021) Capturing heterogeneity among consumers with multitaste preferences. Working paper, New York University Stern School of Business, New York.
- Liu J, Kawaguchi K, Li T (2024) Segmenting consumer location-product preferences for assortment localization. Working paper, Hong Kong University of Science & Technology (HKUST), Hong Kong.

- Liu X, Lee D, Srinivasan K (2019) Large-scale cross-category analysis of consumer review content on sales conversion leveraging deep learning. *J. Marketing Res.* 56(6):918–943.
- Liu J, Toubia O, Hill S (2021) Content-based model of web search behavior: An application to TV show search. *Management Sci.* 67(10):6378–6398.
- Lu Z, Kannan P (2026) AI for customer journeys: A transformer approach. *J. Marketing Res.* 63(1):1–26.
- Malhotra NK, Sudhir K, Toubia O (2023) *Artificial Intelligence in Marketing*, Review of Marketing Research, vol. 20 (Emerald Publishing Limited, Bingley, UK).
- McInnes L, Healy J, Saul N, Großberger L (2018) UMAP: Uniform manifold approximation and projection. *J. Open Source Software* 3(29):861.
- Nie L, Li Y, Feng F, Song X, Wang M, Wang Y (2020) Large-scale question tagging via joint question-topic embedding learning. *ACM Trans. Inform. Systems (TOIS)* 38(2):1–23.
- Ouyang S (2021). Zhihu bets big on China's knowledge-sharing market. Accessed July 29, 2024, <https://global.chinadaily.com.cn/a/202101/14/WS5f5ffde7ca31024ad0baa29fb.html>.
- Puranam D, Kadiyali V, Narayan V (2021) The impact of increase in minimum wages on consumer perceptions of service: A transformer model of online restaurant reviews. *Marketing Sci.* 40(5):985–1004.
- Reimers N, Gurevych I (2019) Sentence-bert: Sentence embeddings using siamese bert-networks. Inui K, Jiang J, Ng V, Wan X, eds. *Proc. EMNLP-IJCNLP* (Association for Computational Linguistics, Stroudsburg, PA), 3982–3992.
- Röder M, Both A, Hinneburg A (2015) Exploring the space of topic coherence measures. Bendersky M, Najork M, Singla A, eds. *Proc. Eighth ACM Internat. Conf. Web Search Data Mining* (Association for Computing Machinery, New York), 399–408.
- Ruiz FJR, Athey S, Blei DM (2020) SHOPPER: A probabilistic model of consumer choice with substitutes and complements. *Ann. Appl. Stat.* 14(1):1–27.
- Schein A, Wallach H, Zhou M (2016) Poisson-gamma dynamical systems. Lee DD, Sugiyama M, von Luxburg U, Guyon I, Garnett R, eds. *Advances in Neural Information Processing Systems*, vol. 29 (Neural Information Processing Systems Foundation, Red Hook, NY).
- Schein A, Paisley J, Blei DM, Wallach H (2015) Bayesian Poisson tensor factorization for inferring multilateral relations from sparse dyadic event counts. Cao L, Zhang C, Li T, Lin T, eds. *Proc. 21th ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining* (Association for Computing Machinery, New York), 1045–1054.
- Schein A, Linderman S, Zhou M, Blei D, Wallach H (2019) Poisson-randomized gamma dynamical systems. *Adv. Neural Inform. Processing Systems*, vol. 32 (Curran Associates Inc., Red Hook, NY).
- Sisodia A, Burnap A, Kumar V (2025) Generative interpretable visual design: Using disentanglement for visual conjoint analysis. *J. Marketing Res.* 62(3):405–428.
- Tian Z, Dew R, Iyengar R (2024) Mega or micro? Influencer selection using follower elasticity. *J. Marketing Res.* 61(3):472–495.
- Toubia O (2021) A poisson factorization topic model for the study of creative documents (and their summaries). *J. Marketing Res.* 58(6):1142–1158.
- Wainwright MJ, Jordan MI (2008) Graphical models, exponential families, and variational inference. *Foundations and Trends® in Mach. Learn.* 1(1–2):1–305.
- Wallach HM, Murray I, Salakhutdinov R, Mimno D (2009) Evaluation methods for topic models. *Proc. 26th Annual Internat. Conf. Machine Learn.*, 1105–1112.
- Wang H, Yeung DY (2020) A survey on Bayesian deep learning. *ACM Comput. Surveys* 53(5):1–37.
- West R, Leskovec J, Potts C (2021) Postmortem memory of public figures in news and social media. *Proc. Natl. Acad. Sci. USA* 118(38):e2106152118.
- Wu X, Dong X, Nguyen TT, Luu AT (2023) Effective neural topic modeling with embedding clustering regularization. *Internat. Conf. Machine Learn.* (PMLR, New York), 37335–37357.
- Yang S, Allenby GM (2003) Modeling interdependent consumer preferences. *J. Marketing Res.* 40(3):282–294.
- Yao E, Zheng G, Jin O, Bao S, Chen K, Su Z, Yu Y (2014) Probabilistic text modeling with orthogonalized topics. Geva S, Trotman A, Bruza P, Clarke CLA, Järvelin K, eds. *Proc. 37th Internat. ACM SIGIR Conf. Res. Development Inform. Retrieval* (Association for Computing Machinery, New York), 907–910.
- Yin M, Zhou M (2018) Semi-implicit variational inference. *Internat. Conf. Machine Learn.* (PMLR, New York), 5660–5669.
- Yin J, Feng Y, Liu Y (2024a) Modeling behavioral dynamics in digital content consumption: An attention-based neural point process approach with applications in video games. *Marketing Sci.* 44(1):220–239.
- Yin M, Gao R, Cong Z (2025) Personalizing language models for generative targeting. Preprint, submitted June 30, <http://dx.doi.org/10.2139/ssrn.5329729>.
- Yin M, Boughanmi K, Mukherjee A, Ansari A (2024b) Meta-learning customer preference dynamics for fast customization on digital platforms. Preprint, submitted March 13, <https://ssrn.com/abstract=4727171>.
- Yu Y, Si X, Hu C, Zhang J (2019) A review of recurrent neural networks: Lstm cells and network architectures. *Neural Comput.* 31(7):1235–1270.
- Zhou M, Chen H, Ren L, Sapiro G, Carin L, Paisley J (2009) Non-parametric bayesian dictionary learning for sparse image representations. *Adv. Neural Inform. Processing Systems*, vol. 22 (Curran Associates Inc., Red Hook, NY).

Mingzhang Yin is an assistant professor of marketing at the Warrenton College of Business, University of Florida. He received a PhD in Statistics from UT Austin and was a postdoctoral researcher at Columbia's Data Science Institute. His research spans machine learning, causal inference, and Bayesian methods for quantitative marketing, with publications in leading journals such as *Marketing Science*, *JASA*, *JMLR*, and in top conference proceedings including *NeurIPS*, *ICML*, *ICLR*, *AAAI*, and *AISTATS*.

Ziwei Cong is an assistant professor of marketing at the McDonough School of Business, Georgetown University. She received her PhD from the Hong Kong University of Science and Technology. Her research focuses on user-generated content and digital platform designs. Her work has been published in *Marketing Science* and *Journal of Marketing Research*. She is the recipient of the 2021 Vithala R. and Saroj V. Rao ISMS Doctoral Dissertation Award and the 2020 Shankar-Spiegel Dissertation Proposal Award.

Jia Liu is an associate professor of marketing and affiliated associate professor in IEDA at HKUST. She received her PhD from Columbia University. Her research spans consumer behavior, big data analytics, recommender systems, revenue management, global branding, and generative AI in business. Her work appears in *Marketing Science*, *Management Science*, *JMR*, and *JM*. She is an MSI Young Scholar and a recipient of the John Little Award and Frank M. Bass Award nomination.